Program : **Diploma in Civil and Rural Engineering**	
Course Code : **4369**	Course Title: **Hydraulics Lab**
Semester : **4**	Credits: **1.5**
Course Category: **Program Core**	
Periods per week: **3 (L:0, T:0, P:3)**	Periods per semester: **45**

Course Objectives:

- To develop understanding of the fundamental principles of fluid mechanics through experimentation

Course Prerequisites:

Topic	Course code	Course name	Semester
Elementary Mathematics		Engineering Mathematics	1
Basics of Engineering Mechanics		Engineering Mechanics	2

Course Outcomes:

On completion of the course, the student will be able to:

COn	Description	Duration (Hours)	Cognitive level
CO1	Verify Bernoulli's Theorem	3	Applying
CO2	Determine coefficients for venturimeter, orifice and notch.	12	Applying
CO3	Familarisation of pipe flow and open channel flow	14	Applying
CO4	Familarise the Hydraulic machines	12	Applying
	Lab Test	4	

CO-PO Mapping:

Course Outcomes	PO1	PO2	PO3	PO4	PO5	PO6	PO7
CO1	3			3			
CO2	3			3			
CO3	3						
CO4	3			3			

3-Strongly mapped, 2-Moderately mapped, 1-Weakly mapped

Course Outline:

Module Outcomes	Description	Duration (Hours)	Cognitive Level
CO1	**Verify Bernoulli's Theorem**		
M1.01	Use Bernoulli's apparatus to apply Bernoulli's theorem to get total energy line for a flow in a closed conduit of varying cross sections	3	Applying
CO2	**Determine coefficients for venturimeter, orifice and notch**		
M2.01	Calibrate Venturimeter to find out the discharge in a pipe.	3	Applying
M2.02	Calibrate the Orifice to find out the discharge through a tank	3	Applying
M2.03	Use triangular notch to measure the discharge through open channel.	3	Applying
M2.04	Use Rectangular notch to measure the discharge through open channel	3	Applying
	Lab Test I	2	
CO3	Familarisation of pipe flow and hydraulic machines		
M3.01	Determine the Darcy's coefficient of friction for pipes, Plot TEL and HEL	3	Applying
M3.02	Study of Hydraulic machines	2	Understanding
M3.03	Determine the efficiency of turbines.	3	Applying
M3.04	Draw the characteristics curves of centrifugal pump and reciprocating pump	3	Applying
M3.05	Determine efficiency of hydraulic ram.	3	Applying

CO4	**Familiarize with local irrigation structures and practices**		
M4.01	Open ended Project	12	Applying
	Lab Test II	2	

Text / Reference:

T/R	Book Title/Author
T1	Dr. R.K.Bansal : Fluid Mechanics & Hydraulic Machine ; Laxmi Publishers
R2	R.S.Khurmi : Hydraulics, Fluid Mechanics & Hydraulic Machines; S. Chand & Co.
R3	Modi &Sethi : Hydraulics & Hydraulic Machines ; Standard Publishers
R4	R.K.Rajput : Hydraulics ; S.Chand& Co.
R5	Jagdish lal : Hydraulics ; Dhanpat Rai & Sons

Online resources:

Sl.No	Website Link
1	https://nptel.ac.in/
2	http://egyankosh.ac.in/ https://www.coursera.org ›
3	https://swayam.gov.in/